香港資優教育學苑
The Hong Kong Academy for Gifted Education

IMO Training 2023 – 24 Phase I
Polynomials (多項式))
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A function (函數) of the form $f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_o$, where n is a positive integer, is called a polynomial in one variable (變元). The numbers $a_n, a_{n-1}, \ldots, a_1, a_o$ are called the coefficients(係數) of the polynomial.
Further, if $a_n \neq 0$, then 'n' is the degree (次) of the polynomial, denoted by deg f.
Two polynomials $f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_o$ and $g(x) = b_mx^m + b_{m-1}x^{m-1} + \ldots + b_1x + b_o$ are equal if and only if m = n and all the corresponding coefficients are equal, and we write $f(x) \equiv g(x)$.

Operations on polynomials (多項式的運算)

Define 1. $f(x) \pm g(x) = (a_n \pm b_n)\, x^n + (a_{n-1} \pm b_{n-1})x^{n-1} + \ldots + (a_1 \pm b_1)x + (a_o \pm b_o)$
2. $k(f(x)) = ka_nx^n + ka_{n-1}x^{n-1} + \ldots + ka_1x + ka_o$
3. $f(x).g(x) = a_nb_mx^{n+m} + (a_{n-1}b_m + a_n b_{m-1})\, x^{n+m-1} + \ldots + (a_1 b_o + b_1 a_o)x + a_o b_o$
4. $f^m(x) = (a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_o)^m$, where m is a positive integer.

Example 1: Let $f(x) = x^4 + x^3 - 3x^2 + x + 2$. Prove that for any positive integer n, the polynomial $f^n(x)$ has at least one negative coefficient.

Example 2: K(k>1) is a given positive integer. Find all polynomials P(x), such that for any real number x, $P(x^k) = P^K(x)$. (Estonia M.O. 2003)

Division Algorithm (多項式的除法)

Suppose deg f(x) ≧ deg g(x), when f(x) is divided by g(x), we obtain the quotient Q(x) and the remainder R(x), where deg R(x) < deg g(x). That is, $f(x) = g(x)Q(x) + R(x)$

Example 3: Divide $5x^3 - 2x^2 + 8$ by $x^2 - 3x + 1$.

Example 4: Divide $3x^4 + 7x^3 - 15x - 20$ by $x + 2$.
(Short Division 短除法)

$$\begin{array}{r|rrrrr} -2 & 3 & 7 & 0 & -15 & -20 \\ & & -6 & -2 & 4 & 22 \\ \hline & 3 & 1 & -2 & -11 & \underline{2} \end{array}$$

The quotient is $3x^3 + x^2 - 2x - 11$ and the remainder is 2

Example 5: Divide $16x^3 + 8x^2 - 2x + 1$ by $2x - 1$.

Example 6: Find the remainder when $x^{80} - 3x^{35} + 2$ is divided by $x^2 - 1$.

Remainder Theorem (餘數定理)

Theorem: The remainder when f(x) is divided by (x-b) is f(b).
Proof: Suppose $f(x) = (x - b)Q(x) + R$, where deg (R) = 0.
Putting $x = b$ gives $f(b) = R$.

Factor Theorem (因式定理)

Given the polynomial $f(x) = a_nx^n + a_{n-1}x^{n-1} + \ldots + a_1x + a_o$, $a_n \neq 0$. The binomial (x-c) is a factor of f(x) if and only if f(c) = 0.

Example 7: Factorize $f(x) = 6x^4 + 13x^3 + 3x^2 + 4x + 4$.

Solution: The possible factors are $x\pm1$, $x\pm2$, $x\pm4$, $2x\pm1$, $3x\pm1$, $3x\pm2$, $6x\pm1$.
Using Factor theorem, x+2 and 3x+2 are factors. Finally $f(x) = (x+2)(3x+2)(2x^2 - x + 1)$

Example 8: Prove that $x(x+1)(2x+1)$ is a factor of $(x+1)^{2m} - x^{2m} - 2x - 1$.

Example 9: Given that a, b, c, d are positive integers, prove that $f(x)=x^{4a} + x^{4b+1} + x^{4c+2} + x^{4d+3}$ is divisible by $x^3 + x^2 + x + 1$.

Example 10: Prove that $(a+b)(b+c)(c+a) + abc = (a+b+c)(ab + bc + ca)$
Example 11: Prove that $(a+b+c)^3 - a^3 - b^3 - c^3 = 3(a+b)(b+c)(c+a)$

Factorization of Homogeneous Polynomials (齊次多項式)
Example 12: Factorize $P(a, b, c) = a^3 + b^3 + c^3 - 3abc$
Example 13: Factorize $F(a, b, c) = \Sigma(b-c)(b+c)^3$.
Example 14: Factorize $F(a, b, c) = (a+b+c)^5 - a^5 - b^5 - c^5$.
Example 15: Factorize $F(a, b, c, x) = \sum_{abc}(a-b)^3(x-c)^2$
Example 16: Find all polynomials with real coefficients satisfying
$(x^3 + 3x^2 + 3x + 2)P(x-1) = (x^3 - 3x^2 + 3x - 2) P(x)$. (Vietnam MO 2003)
Example 17: (45th IMO 2004 No. 2) Find all polynomials f with real coefficients given that for all real numbers a, b, c such that $ab+bc+ca = 0$ we have the following relationship $f(a-b) + f(b-c) + f(c-a) = 2f(a+b+c)$.

Eisenstein Criterion (愛森斯坦判別法)
(Named after Gotthold Eisenstein, 1823-1852, a German Mathematician)
If $f(x) = a_nx^n + a_{n-1}x^{n-1} + \dots + a_1x + a_o$, where $a_o, a_1, a_2, \dots, a_n$ are all integers, and there exists a prime number p such that (i) $p|a_i$ for $i = 0, 1, 2, \dots, n-1$, (ii) $p \nmid a_n$, (iii) $p^2 \nmid a_o$, then f(x) is not reducible over the set of rational numbers. (不可約多項式).

Proof: Assume that there exist g(x) and h(x) such that $f(x) = g(x)h(x)$, and
$g(x) = b_rx^r + b_{r-1}x^{r-1} + \dots + b_1x + b_o$, $h(x) = c_sx^s + c_{s-1}x^{s-1} + \dots + c_1x + c_o$, $r, s \in N^+$

We have $a_o = b_oc_o$. Since $p|a_o$ and $p^2 \nmid a_o$, hence w.l.o.g. we can assume that $p|b_o$ and $p \nmid c_o$.

On the other hand, since $p \nmid a_n$, and $a_n = b_rc_s$, hence $p \nmid b_r$,

Now consider $b_o, b_1, \dots, b_{r-1}, b_r$, starting from b_o, we have the coefficients divisible by p, until we come to a certain b_k, where $0<k<r$.

(i.e., $p|b_o$, $p|b_1$, $p|b_2$, $\dots p|b_{k-1}$, $p \nmid b_k$, $p \nmid b_{k+1}, \dots, p \nmid b_r$)

Next consider $a_k = b_kc_o + b_{k-1}c_1 + \dots + b_oc_k$
Every term on RHS is a multiple of p except the first term, hence a_k is not divisible by p.
This is contradictory to condition (i).
Hence the assumption is wrong, and f(x) is irreducible.

Example 18: The polynomial $f(x) = 2x^4 - 5x^3 + 10x + 15$ is irreducible.
Example 19: Prove that the polynomial $f(x) = 2x^4 + 9x^2 + 6$ is irreducible. (Take p=3)
Example 20: The polynomial $f(x) = x^7 - 48x^2 + 24$ is irreducible. (Take p = 3)
Example 21: Prove that he polynomial $f(x) = x^2 - k$, where k is a prime number, is irreducible. (Take p = k)
Example 22: Prove that he polynomial $f(x) = x^2 + x + 2$ is irreducible.
(Hint: Put $x = y+3$ to change the polynomial to $y^2 + 7y + 14$, and take p=7.)
Example 23: Prove that he polynomial $f(x) = x^4 + 1$ is irreducible.
Example 24 If p is a prime number, prove that the polynomial
$f(x) = x^{p-1} + x^{p-2} + x^{p-3} + \dots + x + 1$ is irreducible.
Example 25: Given that $f(x) = a_{2003}x^{2003} + a_{2002}x^{2002} + \dots\dots + a_1x + a_o$, anf that there exist positive integers p, q, r, where $p<q<r$, satisfying $f(p) = q$, $f(q) = r$, and $f(r)=p$. Prove that among the coefficients of f(x), at least one of them is not an integer.
(Hungarian MO 2002)

Euclidean Algorithm(輾轉相除)

If f(x) and g(x) are polynomials of degrees n and m, both greater than 1. Suppose there exists a polynomial h(x), unique apart from a constant factor, such that h(x) is the Greatest Common Divisor of f(x) and g(x), then h(x) can be found using the Method of Euclidean Algorithm, as follows:

Method: Given the polynomials f(x) and g(x), with all common constant factors removed beforehand.

Suppose $\deg f(x) \geqq \deg g(x)$, then $f(x) = g(x)\, q_1(x) + r_1(x)$ ----------------(1)

$$
\begin{aligned}
g(x) &= r_1(x)\, q_2(x) + r_2(x) \\
r_1(x) &= r_2(x)\, q_3(x) + r_3(x) \\
r_2(x) &= r_3(x)\, q_4(x) + r_4(x) \\
&\ldots\ldots \\
r_{n-2}(x) &= r_{n-1}(x)\, q_n(x) + r_n(x) \\
r_{n-1}(x) &= r_n(x)\, q_{n+1}(x)
\end{aligned}
$$

Finally, $r_n(x)$ is the G.C.D. of f(x) and g(x).

Proof: Consider (1). Let d(x) be the Greatest Common Divisor (G.C.D.) of f(x) and g(x) and d'(x) be the G.C.D. of g(x) and $r_1(x)$.
Since d'(x) be the G.C.D. of g(x) and $r_1(x)$, then it is a factor of f(x), and hence d(x).
Similarly by rewriting in the form $r_1(x) = g(x)\, q_1(x) - f(x)$, we see that since d(x) is a factor of both f(x) and g(x), this means that d(x) is a factor of $r_1(x)$, and hence a common factor of $r_1(x)$ and g(x) and hence a factor of d'(x).
From above we see that d(x) and d'(x) are factors of each other and they differ by at most a constant.
As common constant factors have been removed beforehand, therefore the G.C.D. of f(x) and g(x) is the same as the G.C.D. of g(x) and $r_1(x)$.
Repeating the above procedure, we can say that the G.C.D. of f(x) and g(x), g(x) and $r_1(x)$, $r_1(x)$ and $r_2(x)$, $r_2(x)$ and $r_3(x)$, …, $r_{n-1}(x)$ and $r_n(x)$ are all the same.
Now $r_n(x)$ is a factor of $r_{n-1}(x)$.
Hence $r_n(x)$ is the G.C.D. of $r_{n-1}(x)$ and $r_n(x)$, and hence f(x) and g(x).

Example 26: Find the G.C.D.of $f(x) = 2x^4 - x^3 - 5x^2 + 13x - 12$ and $g(x) = 6x^4 - 5x^3 + 17x - 12$.

Example 27: Find the G.C.D. of $f(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$ and $g(x) = x^3 - x^2 - 4x + 4$.

Bézout's Theorem for Polynomials (裴蜀定理)

(Named after Etienne Bezout, 1730-1783, a French Mathematician.)

If f(x) and g(x) are two polynomials with the G.C.D. h(x), then there exist two other polynomials F(X) and G(X) such that $f(x)F(x) + g(x)G(x) = h(x)$.
By suitable choice we can always have $\deg F(x) < \deg g(x)$ and $\deg G(x) < \deg f(x)$.
For if $\deg F(x) > \deg g(x)$, then we can choose $\lambda(x)$ such that $F(x) - \lambda(x)g(x)$ is of degree less than g(x), and obtain $f(x)[F(x) - \lambda(x)g(x)] + g(x)[G(x) + \lambda(x)f(x)] = 1$.
Here we have $F'(x) = [F(x) - \lambda(x)g(x)]$ and $\deg F'(x) < \deg g(x)$, and

$G'(x) = [G(x) + \lambda(x)f(x)]$ and $\deg G'(x) < \deg f(x)$.

In particular, if f(x) and g(x) are co-prime, then there exist polynomials F(x) and G(x) such that $f(x)F(x) + g(x)G(x) = 1$

Theorem: For any co-prime polynomials f(x) and g(x), if F(x) and G(x) are so chosen that $f(x)F(x) + g(x)G(x) = 1$, and deg F(x) < deg g(x), deg G(x) < deg f(x), then F(x) and G(x) are unique.

Proof: Suppose F(x), G(x) and $F_1(x)$, G(x) satisfy the given conditions, that is
$f(x)F(x) + g(x)G(x) = 1$ and $f(x)F_1(x) + g(x)G_1(x) = 1$, and
deg F(x) < deg g(x) and deg G(x) < deg f(x), deg $F_1(x)$ < deg g(x) and deg $G_1(x)$ < deg f(x).
then $[F(x) – F_1(x)]f(x) = [G_1(x) – G(x)]g(x)$.
Now f(x) and g(x) are co-prime, hence f(x) is a factor of $[G_1(x) – G(x)]$.
However, deg $[G_1(x) – G(x)]$ < deg f(x).
This leads to contradictions.
The only possible case is that $G_1(x) = G(x)$, and similarly $F_1(x) = F(x)$.
That is, the existence of F(x) and G(x) are unique.

Example 28: Find the G.C.D. of $f(x) = x^4 +3x^3 + x^2 -2$ and $g(x) = x^4 – x^3 + x^2 + 2$, hence express the G.C.D in the form $f(x)F(x) + g(x)G(x)$.

Example 29: Show that $f(x) = x^3 + x + 1$ and $g(x) = x^2 – x + 1$ are co-prime, and hence express 1 in the form $f(x)F(x) + g(x)G(x)$ for some polynomials F(x) and G(x).

Example 30: Find a polynomial p(x) such that (x^2+1) is a factor of p(x), and $x^3 + x^2 + 1$ is a factor of $p(x) + 1$.

Root of a Polynomial (多項式的根)

Given a polynomial f(x). If $f(\alpha) = 0$, then α is called a zero or a root of the polynomial.

Fundamental Theorem of Algebra (代數基本定理/高斯定理)

Every non-constant single variable polynomial has at least one root, which may be real or complex.

Theorem: A polynomial of degree n (n≧1) has exactly n roots, though some of the roots may be repeated roots, and the roots may be real or complex.

Proof: Let the polynomial be f(x).

Part 1: Using the Fundamental Theorem of algebra, it has at least one root, say r_1.
Then $f(x) = (x – r_1) g(x)$, where g(x) is a polynomial of degree n-1.
Suppose $n-1\geq 1$, then again it has at least one root, say r_2.
So, $f(x) = (x – r_1)(x – r_2)h(x)$
The process may be repeated until we obtain $f(x) = a_n(x – r_1)(x – r_2)\ldots(x – r_n)$, where a_n is the coefficient of x_n in f(x).
Hence f(x) has n roots

Note: If a root occurs twice, it is called a double root. If it occurs three times, it is called a triple root. If it occurs m times, it is called a root of multiplicity m.(m 多重根)

Part 2: Suppose a polynomial f(x) of degree n has m roots, where m>n.
Now, take any n roots $r_1, r_2, \ldots, r_n$, we have
$f(x) = a_n (x – r_1)(x – r_2)\ldots(x – r_n)$.
Further, f(x) is zero for at least one other root, say r_{n+1}.
Hence $f(r_{n+1}) = a_n (r_{n+1} – r_1)(r_{n+1} – r_2)\ldots(r_{n+1} – r_n)$
We must have $a_n = 0$, and hence f(x) is not a polynomial of degree n..
This is contradictory.
Hence the assumption is wrong, and f(x) has at most n roots.

Corollary: If f(x) and g(x) are polynomials of degree n, and $f(x_i) = g(x_i)$ with x_1, $x_2, \dots, x_m$ distinct numbers and m>n, then $f(x) \equiv g(x)$.

Proof: f(x) – g(x) is a polynomial of degree less than or equal to n, but with m distinct roots, where m>n.

Using the above Theorem, $f(x) - g(x) \equiv 0$, hence $f(x) \equiv g(x)$.

Example 31 Given that a, b and c are distinct real numbers, prove that

(a) $$\frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} + \frac{(x-a)(x-b)}{(c-a)(c-b)} = 1$$

(b) $$\frac{a^2(x-b)(x-c)}{(a-b)(a-c)} + \frac{b^2(x-c)(x-a)}{(b-c)(b-a)} + \frac{c^2(x-a)(x-b)}{(c-a)(c-b)} = x^2 .$$

Example 32: Given that $a_1, a_2, .., an, b_1, b_2, \dots,, b_n$ are 2n distinct real numbers, solve the following system of equations:

$$\frac{x_1}{a_1-b_1} + \frac{x_2}{a_1-b_2} + \dots + \frac{x_n}{a_1-b_n} = 1,$$
$$\frac{x_1}{a_2-b_1} + \frac{x_2}{a_2-b_2} + \dots + \frac{x_n}{a_2-b_n} = 1,$$
$$\dots\dots$$
$$\frac{x_1}{a_n-b_1} + \frac{x_2}{a_n-b_2} + \dots + \frac{x_n}{a_n-b_n} = 1,$$

Example 33: Prove that $(x+1)(x+2)(x+3)(x+4) + 1 = (x^2 + 5x + 5)^2$.

Example 34: Prove that $f(x) = (x-1)(x-2)(x-3)\dots\dots(x-10) - 1$ is irreducible over the set of integers

Example 35: Let f(x) be a polynomial of degree n and with integral coefficients. Prove that if there are at least 2n+1 distinct integers m such that |f(m)| is a prime number, then f(x) is irreducible over the set of integers.

Method of Equating Coefficients

Example 36: Find the values of a and b so that $f(x) = x^4 + ax^3 + bx^2 - 8x + 4$ is the square of a polynomial.

Example 37: Find the sum to n terms of the series $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$

Example 38: Find the sum to n terms of the series $1^4 + 2^4 + 3^4 + \dots + n^4$

Example 39: Find the sum to n terms of the series $1^3 + 3^3 + 5^3 + \dots$

Example 40: Find all polynomials P, in two variables, with the following properties:

(i) For a positive integer n ad all real t, x, y, $P(tx, ty) = t^n P(x, y)$. (that is, P is a homogeneous of degree n.)

(ii) For all real a, b, c, P(b+c, a) + P(c+a, b) + P(a+b, c) = 0.

(iii) P(1, 0) = 1.

(17^{th} IMO 1975 Number 6.)